



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

Lab experiment-9

Travelling Salesman Problem (TSP)

Aim

To write a Python program to implement the Travelling Salesman Problem (TSP).

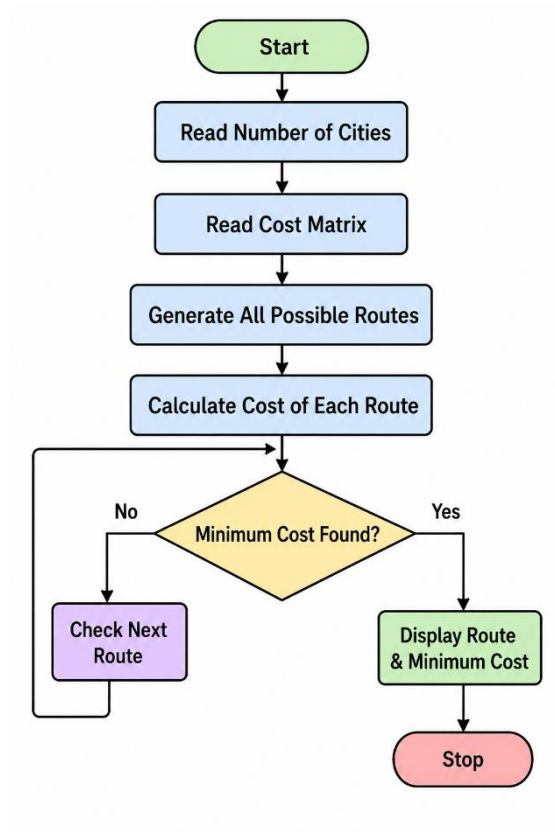
Objective

To find the minimum cost path that visits every city exactly once and returns to the starting city.

Algorithm

1. Start.
2. Read the number of cities.
3. Read the cost (distance) matrix.
4. Generate all possible routes starting from the first city.
5. Calculate the total cost of each route.
6. Select the route with the minimum cost.
7. Display the optimal route and its cost.
8. Stop.

Flowchart



Code

```
from itertools import permutations

n = int(input("Enter number of cities: "))

cost = []

print("Enter the cost matrix:")

for i in range(n):
    cost.append(list(map(int, input().split())))

cities = list(range(n))

min_cost = float('inf')

best_path = ()

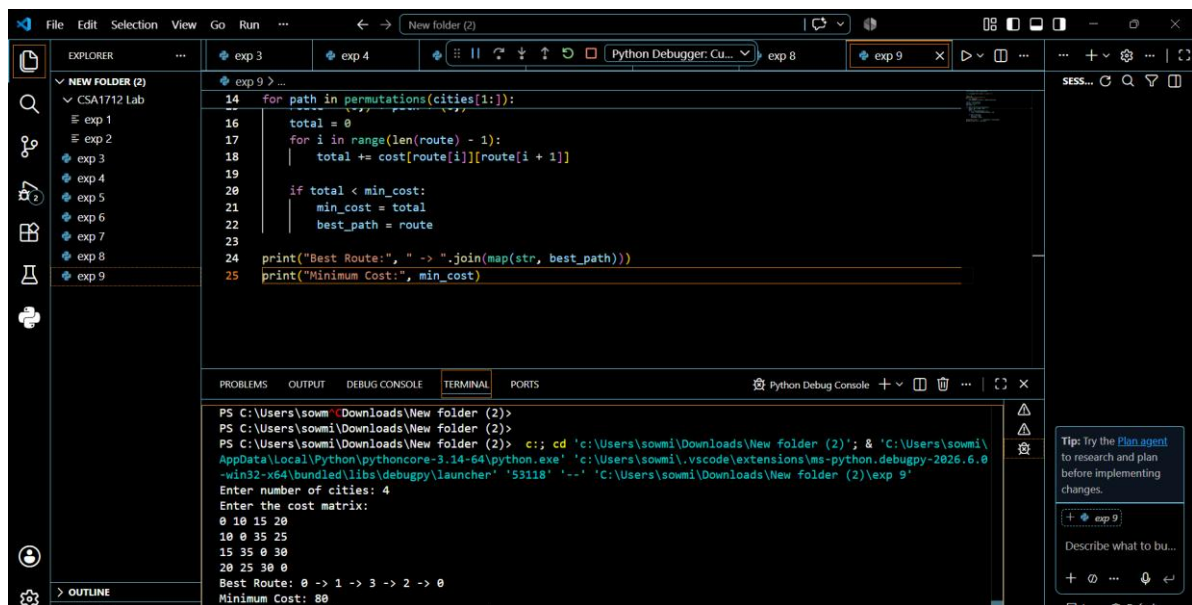
for path in permutations(cities[1:]):
    route = (0,) + path + (0,)
    total = 0
```

```

for i in range(len(route) - 1):
    total += cost[route[i]][route[i + 1]]
if total < min_cost:
    min_cost = total
    best_path = route
print("Best Route:", " -> ".join(map(str, best_path)))
print("Minimum Cost:", min_cost)

```

Output



```

14 for path in permutations(cities[1:]):
15     total = 0
16     for i in range(len(route) - 1):
17         total += cost[route[i]][route[i + 1]]
18
19     if total < min_cost:
20         min_cost = total
21         best_path = route
22
23 print("Best Route:", " -> ".join(map(str, best_path)))
24 print("Minimum Cost:", min_cost)

```

```

PS C:\Users\sowmi\Downloads\New folder (2)>
PS C:\Users\sowmi\Downloads\New folder (2)>
PS C:\Users\sowmi\Downloads\New folder (2)> c:; cd 'C:\Users\sowmi\Downloads\New folder (2)'; & 'C:\Users\sowmi\AppData\Local\Python\pythoncore-3.14-64\python.exe' 'c:\Users\sowmi\.vscode\extensions\ms-python.debugpy-2026.6.0-win32-x64\bundle\libs\debugpy\launcher' '53118' '--' 'C:\Users\sowmi\Downloads\New folder (2)\exp 9'
Enter number of cities: 4
Enter the cost matrix:
0 10 15 20
10 0 35 25
15 35 0 30
20 25 30 0
Best Route: 0 -> 1 -> 3 -> 2 -> 0
Minimum Cost: 80

```

Result

The optimal route and minimum travel cost were obtained successfully.

Conclusion

The Travelling Salesman Problem was solved successfully by finding the minimum-cost route that visits every city exactly once and returns to the starting city.